

A Tool for Certifying Exponent Thresholds as Exact Linear-Programming Facets

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Statement of Purpose

This note presents a tool, not a theorem. The tool, `certify.py`, makes a published exponent threshold an executable object. A problem instance is a data file declaring block variables, slice parameters, decomposition pieces, and tools encoded as negated admissibility conditions, and the engine computes exact optima over $\mathbb{Q}$ by exhaustive vertex enumeration, with no floating point anywhere. The tool serves three purposes. The first is reproduction: a run counts as a certificate only when its output matches expected values declared from the source publication, and every run ends in an explicit verdict. The second is audit: a mismatch is either a transcription error or a genuine discrepancy in the source, and the tool localizes it to a slice and a facet. The third is acceptance testing: a candidate estimate is one more data stanza, and a rerun reports in seconds whether the threshold moves and which constraint binds next. We demonstrate all three purposes on two instances. One reproduces the threshold profile of a conditional reduction for primes in the intervals $[4n^2 - n, 4n^2 + n]$. The other transcribes Rivat and Sargos’s theorem on Piatetski-Shapiro primes and certifies three published rationals from a single atom table: $c < 2817/2426$, $c < 441/380$, and the prior record $c < 15/13$, the latter two recovered as tool-removal ablations, each with its witness on the predicted facet. We also state what certificates do not establish, using the companion paper of Rivat and Wu as the boundary case: its exponential-sum layer is certifiable, while its sieve positivity layer, resting on numerical integration of Buchstab functions, is not.

1 Introduction

A recurring pattern in analytic number theory is the exponent record: a rational number such as $2817/2426$ or $243/205$ marking how far a method reaches, produced by balancing finitely many estimates over the ranges of grouped variables. In the cleanest cases the record is literally the optimum of a finite system of linear inequalities with rational coefficients. The optimum is stated in the paper, and the system is scattered through it: one window condition per lemma, one cap per error term, a design choice fixing a parameter here, a combinatorial identity imposing a ceiling there. Verifying the record means reassembling that system and checking the arithmetic, which referees do by hand, once.

This note describes a tool that makes the reassembled system a first-class, executable object. The engine, `certify.py`, is problem-agnostic and fits in a few hundred lines. Everything problem-specific lives in a JSON instance file: variables, parameters, decomposition pieces, and tools, with every coefficient an exact rational. The engine enumerates the polytopes on which every

tool fails simultaneously and computes exact optima by vertex enumeration over $\mathbb{Q}$. Its output is compared against expected values declared in the instance and taken from the source publication, and the run ends with a verdict. We call a passing run an *exponent certificate* (Definition 2.3). The same machinery answers a second question at essentially no cost: given a hypothetical new estimate, encoded as one more tool, does the threshold move, and where does the binding constraint migrate? A published record thus acquires an executable acceptance test for candidate improvements.

Two instances accompany the tool. The first certifies the threshold profile of a conditional reduction for primes in square-centered intervals $[1, 2]$, reproducing the profile $\zeta(\gamma) = (6 - 5\gamma)/4$, its witness, and a tool-ablation byproduct. The second transcribes Rivat and Sargos [3] and is the better advertisement for the method: a single atom table certifies three published rationals, two of them as ablations, all attained on the predicted facet (Section 5). The companion paper of Rivat and Wu [4] supplies the boundary of scope (Section 7).

2 The certification framework

Definition 2.1 (Instance). *An instance consists of block variables $x = (x_1, \dots, x_n)$, slice parameters $\theta = (\theta_1, \dots, \theta_k)$, a designated objective coordinate x_j , a finite set of tools, a finite set of pieces, a finite list of rational slice values, and declared expected values. A constraint is an inequality $a \cdot x \leq b_0 + b \cdot \theta$ with $a \in \mathbb{Q}^n$, $b_0 \in \mathbb{Q}$, $b \in \mathbb{Q}^k$. A tool is the negation of an estimate’s admissibility condition, written in disjunctive normal form: a list of fail options, each a conjunction of constraints, the tool failing at x when at least one option holds. A piece is a pair (region, applicable tools), the region being a bounded rational-linear set of block coordinates that the piece must cover.*

Fix a slice θ . A *total-failure witness* in a piece is a point of its region at which every applicable tool fails. The engine finds witnesses by distributing the disjunctions: for each choice of one fail option per tool it forms the polytope cut out by the region together with the chosen fail atoms, and it examines every vertex, solving each n -subset of constraint rows exactly over $\mathbb{Q}$. Two certified statement types are supported.

Minimization mode. The certified object is $\zeta(\theta) = \min x_j$ over all witnesses at slice θ : the first level at which total failure occurs. The expected value is a closed form, affine in θ with an optional floor.

Coverage mode. The certified object is witness existence per slice. The expected value is a threshold θ^* together with the assertion that a witness exists exactly for $\theta \leq \theta^*$.

Convention 2.2 (Facet convention). *All inequalities are closed, and the ε , η , κ margins of the source arguments are dropped. Certified thresholds are therefore attained: at the critical slice the witness set collapses onto the binding facet, and the engine reports the facet point. Reported thresholds are the suprema of failing slices, matching the usual convention that the analytic result holds on the open side.*

Definition 2.3 (Exponent certificate). *An exponent certificate for a published threshold is a run of the engine on a declared instance whose computed values match, exactly and in $\mathbb{Q}$, expected values declared in the instance and taken from the source publication, for every slice and every declared ablation.*

The match requirement is what separates certification from computation. A run with no declared expected values computes a profile but certifies nothing, and the engine says so (Section 3).

3 The soundness contract

The engine certifies the arithmetic of the encoded system exactly. It does not certify that the encoding is faithful to the source paper. Fidelity is an obligation on the instance author, discharged in the instance file itself, which carries per-tool provenance to numbered statements of the source and a documented list of encoding decisions. The discipline follows the sandwich pattern of [2]. Deliberate relaxations are recorded with their direction: omitting a tool condition can only make failure easier, so certified thresholds are conservative, and a hand-proved coverage argument closes the gap on the relevant region. Design choices of the source, such as a parameter the paper fixes rather than quantifies, are recorded as substitutions, not variables. Window edges that the source covers by construction, through η -shifted endpoints, carry no fail atom; only genuine caps do.

A run ends in one of three verdicts. `CERTIFICATE PASSED` means every computed value matches its declared expectation, which is the reproduction of a published result. `CERTIFICATE FAILED` means at least one mismatch, which is an audit finding: either a transcription error, to be fixed and rerun, or a genuine discrepancy with the source, to be investigated by hand. `NOTHING CERTIFIED` means the instance declared no expected values, which is the exploration regime: the acceptance-test workflow of Section 6 produces predictions awaiting theorems, not checks against them. Keeping the third verdict distinct prevents an exploratory run from masquerading as a certificate.

The transcription of an instance from a paper is mechanical enough to delegate, to a careful reader or to an AI assistant working under the protocol distributed with the tool [15], because the acceptance test is self-checking. An unfaithful atom table has no mechanism for reproducing the source’s exact published rationals, let alone several of them simultaneously under ablation. The judgment calls concentrate in two places: distinguishing genuine caps from design choices, and detecting where the source leaves the rational-linear regime. Those calls are audited by a human against the paper. The arithmetic audits itself.

4 Instance A: a square-interval tail decomposition

The first instance transcribes the constraint system of [1], whose companion inventory [2] documents the encoding in full. The block variables are three exponents (μ, η, β) of a bilinear decomposition, the slice parameter is γ , and the toolkit comprises the Robert–Sargos trilinear estimate [9] in four groupings, the Fouvry–Iwaniec monomial estimate [10] in three, and an elementary tool combining Kuzmin–Landau with the exponent pairs $(1/2, 1/2)$ and $(1/6, 2/3)$. Two pieces carry the type I and type II ranges.

In minimization mode the engine reproduces the declared profile $\zeta(\gamma) = (6 - 5\gamma)/4$ on $\gamma \in [1, 6/5]$ at six slices, exactly. At $\gamma = 11/10$ the reported witness is $(\mu, \eta, \beta) = (11/20, 9/40, 1/8)$, on the facet $4\beta + 5\gamma = 6$, with the binding fail atom of each tool named in the output, matching the inventory’s hand-derived witness. A declared ablation removes the three Fouvry–Iwaniec groupings and certifies that the profile is unchanged, reproducing the inventory’s byproduct that

the coverage argument does not need that estimate. The full run, profile plus ablation, takes about four seconds.

5 Instance B: Rivat–Sargos, three rationals from one table

Rivat and Sargos [3] prove $\pi_c(x) \sim x/(c \log x)$ for $1 < c < 2817/2426$, where $\pi_c(x)$ counts $n \leq x$ with $\lfloor n^c \rfloor$ prime. Writing $\gamma = 1/c$ and $\nu = \log_x N$ for the length of the distinguished range, their argument runs Heath-Brown’s identity [5] with $\ell = 5$ and the design choice $\alpha = 1 - \gamma$, splitting the required coverage into a type II window, a type I requirement, and a residual type I’ window $(2(1 - \gamma), 1/3]$ on which the paper’s new estimate acts. Every condition in the critical path is rational-affine in (ν, γ) . The type II tool of Heath-Brown [6] caps at $\nu \leq 5\gamma - 4$. The thesis type I bound requires $\nu \geq 3(1 - \gamma)$. The new estimate, at Théorème 1 strength, caps at $\nu \leq (939\gamma - 761)/143$ together with $\nu \leq \gamma - 1/2$, $\nu \leq 3\gamma - 2$, and $\nu \leq (5\gamma + 9)/33$, and at Théorème 2 strength, before the improved fifth-derivative criterion of Sargos [7], at $\nu \leq (147\gamma - 119)/23$ and $\nu \leq (597\gamma - 469)/133$ in place of the first and last caps.

The instance runs in coverage mode with both strengths present as separate tools and two declared ablations. The certified results are collected in Table 1. In each row the threshold is an exact rational, the witness at threshold sits at $\nu = 1/3$ exactly, and the threshold matches a rational stated in the source. The base toolkit certifies Théorème 1. Removing the Théorème 1 tool certifies Théorème 2. Removing both new tools leaves Heath-Brown covering type I’ up to $\nu = 5\gamma - 4$, and the certified threshold is $\gamma^* = 13/15$, that is $c < 15/13$: precisely the prior record of Liu and Rivat [8] that the paper identifies with the trivial estimate of its central sum. Eighteen slice checks pass, and the run takes under a second.

Toolkit	Threshold γ^*	Statement	Source claim
Full	2426/2817	$c < 2817/2426$	Théorème 1
Remove T1 tool	380/441	$c < 441/380$	Théorème 2
Remove both new tools	13/15	$c < 15/13$	Liu–Rivat baseline

Table 1: Certified thresholds for the Rivat–Sargos instance. Each is an exact facet of the declared linear system, with the witness at threshold at $\nu = 1/3$. The two ablation rows are the paper’s own internal structure, recovered by tool removal.

The table settles a small point of record. The secondary literature transmits the Rivat–Sargos threshold in two versions, 2817/2426 and 2817/2425. The paper states 2817/2426, and the certificate confirms that 2817/2426 is the exact facet of the paper’s own system: at $\gamma = 2426/2817$ the binding cap satisfies $(939\gamma - 761)/143 = 1/3$ identically.

6 The acceptance-test workflow

Because tools are data, a candidate estimate is one more JSON stanza. Adjoining it and rerunning answers, in seconds and in exact arithmetic, whether the candidate moves the threshold and which constraint binds next. In an experiment on Instance A, a fabricated strengthening of one Robert–Sargos window lifted the profile at every interior slice and migrated the binding piece from type II to the elementary type I facet, locating the next obstruction without any analysis. Such runs end in the third verdict, NOTHING CERTIFIED, and their outputs are predictions conditional

on the candidate becoming a theorem. The workflow inverts the usual economics of checking: the months-long question “would estimate X improve result Y ” becomes a transcription plus a run, and the published record ships with a machine-checkable specification of what any improvement must supply.

7 What is not certified

Certification decides membership in the reachable set of a declared toolkit, exactly, and decides nothing beyond it. No enumeration over type I and type II atoms can manufacture a new atom, and for prime-detection problems there is a principled obstruction behind this: measures agreeing on all such data may disagree about primes, which is the parity phenomenon, circumvented to date only by estimates exploiting structure specific to a sequence [11]. A certificate that a threshold is optimal for its toolkit is a certified impossibility for rearrangement, and simultaneously a specification, through its active facet, of the estimate that any crossing must supply.

The scope boundary is visible within a single problem and a single year. Rivat and Wu [4] prove $\pi_c(x) \gg x/(c \log x)$ for $c \leq 243/205$. Their exponential-sum layer is certifiable in exactly the present sense: the windows of their Corollaries 1 and 2, with endpoints $3 - 3\gamma$, $(61\gamma - 49)/11$, $(60 - 61\gamma)/11$, and $3\gamma - 2$, are rational-affine. Their sieve layer is not. The threshold $243/205$ is the point at which a function $F(\lambda)$, a sum of two- to four-dimensional integrals of the Buchstab function over simplex domains, remains provably positive under symbolic and numerical integration in Maple, with domains manually enlarged for tractability. That is a different kind of computation, with different verification needs, and an instance file must not silently span the boundary between the two. The correct posture is to certify the linear layer and state where it ends.

For the same reason, this tool is not formal verification. A certificate checks that a declared linear system reproduces published rationals under exact arithmetic. It does not verify the analysis justifying the atoms, which is what a proof assistant would demand. The contract here is deliberately lighter, and the per-atom provenance required of instances is the bridge for anyone who wants to carry a certified system further.

8 Related work

The nearest neighbors are the systematizations of exponent information: the analytic number theory exponent database of Tao, Trudgian, and Yang [12] and the automated optimization of exponent pairs [13], which derive records by searching within known estimates, and the Polymath 8 project [14], which optimized a fixed argument’s numerology at scale. The present tool is complementary to these: it certifies published thresholds as exact facets of declared systems, recovers a paper’s internal theorem structure by ablation, and attaches an executable acceptance test to a record. Instances are small data files, so certified systems could in principle be contributed to, or checked against, a database of the kind just cited.

9 Availability

The engine, the schema with the transcription protocol, and both instance files are available at [15]. The engine is a single Python file with no dependencies beyond the standard library,

using `fractions.Fraction` throughout. Runs reported here complete in seconds on commodity hardware. A frozen release with a DOI will accompany this note.

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